



Article

“Artificial Intelligence in Education: An Empirical Analysis of Its Influence on Students’ Engagement”

Article History

Name Of Author:

SRIDEVI M¹

Affiliation:

¹Assistant Professor, Department of Commerce and Management, Vijaya College, Jayanagar, Bengaluru

Corresponding Author:

SRIDEVI M¹

Received:03/07/2026

Revised: 09/07/2026

Accepted: 15/07/2026

Published: 17/07/2026

©2026 the Author(s). This is an open access article distributed under the terms of the Creative Commons Attribution License

<http://creativecommons.org/licenses/by/4.0>

Abstract

The integration of artificial intelligence (AI) in education has transformed traditional learning environments by enhancing accessibility, personalization, and efficiency. This study aims to empirically analyze the influence of AI on students’ engagement and learning outcomes in Bengaluru. Using primary data collected from students through a structured questionnaire, the study evaluates how AI-based tools impact academic interaction, participation, and performance. The findings indicate that AI significantly enhances student engagement and improves learning outcomes, although certain challenges such as overdependence and reduced critical thinking were observed. The study contributes to understanding the role of AI in shaping modern educational practices.

Keywords: Artificial Intelligence, Student Engagement, Learning Outcomes, Digital Learning, Education Technology.



1. Introduction

The rapid advancement of artificial intelligence (AI) has brought significant changes across industries, including education. AI-powered tools such as chatbots, personalized learning platforms, and recommendation systems have redefined the teaching-learning process. These tools enable students to access customized content, receive instant feedback, and improve their understanding of complex concepts.

In a technologically advanced city like Bengaluru, where digital adoption is high, students are increasingly relying on AI tools for academic purposes. However, while AI offers numerous advantages, it also raises concerns regarding dependency and its impact on critical thinking abilities.

Therefore, it becomes essential to empirically examine how AI influences students' engagement and learning outcomes.

2. Literature Review

Recent literature highlights the accelerating transformation of workforce skill requirements in response to technological advancements, particularly artificial intelligence (AI).

According to the World Economic Forum (2022), by 2030, demand for technological skills is expected to grow by 55%, while demand for social and emotional skills will increase by 24%, and physical/manual skills will decline by 14%. This shift challenges the traditional separation between technical and soft skills, positioning them instead as complementary competencies essential for future employability.

Building on this, Manyika et al. (2023) introduced the concept of "skill adjacencies," referring to the ability of individuals to transition between roles by leveraging existing skill sets with minimal retraining. Their findings suggest that targeted reskilling through adjacency mapping could address up to 60% of projected skill gaps, making it a cost-effective workforce strategy.

Similarly, Singh and Dey (2022) identified critical thinking, digital fluency, and collaborative problem-solving as core competencies required across high-growth sectors, particularly in the context of India's evolving economic landscape under NEP 2020.



The integration of AI further strengthens these evolving skill demands. Studies indicate that AI-driven tools enhance learning efficiency, adaptability, and decision-making capabilities among individuals.

AI-enabled systems support personalized learning, real-time feedback, and continuous skill development, thereby improving both productivity and innovation outcomes. Moreover, organizations adopting AI-integrated learning ecosystems report significantly higher employee engagement and performance levels.

Gratton (2023) argues that organizations must shift from traditional employee-employer relationships to collaborative learning ecosystems that emphasize continuous upskilling.

Supporting this, Chakraborty and Misra (2024) found that organizations with structured learning cultures experienced 2.3 times higher employee retention and 41% greater innovation performance, demonstrating the tangible benefits of investing in workforce skill development.

Overall, the literature underscores that the future workforce will require a hybrid skill set combining technical expertise, cognitive abilities, and interpersonal competencies, with AI acting as a key enabler of this transformation.

3. Research Gap

Most existing studies on artificial intelligence (AI) in education rely heavily on secondary data, conceptual frameworks, or global datasets, with limited use of primary evidence from students. This creates a gap between theoretical findings and actual student experiences.

There is also a lack of region-specific research, especially in the Indian context. Cities like Bengaluru, despite being major technology and education hubs, remain underexplored, limiting the relevance of existing findings.

Although prior research highlights AI's potential to improve student engagement and learning outcomes, results are often inconsistent, particularly in measuring its impact on academic performance and engagement levels.



Many studies also examine engagement and learning outcomes separately rather than analyzing their relationship in AI-based learning environments.

Furthermore, limited attention is given to students' perceptions and interactions with AI tools, including usability, dependency, and effectiveness.

These gaps highlight the need for an empirical, region-specific study.

4. Objectives of the Study

1. To study how AI affects students' engagement in learning in educational institutions in Bengaluru.
2. To understand how AI impacts students' learning outcomes, such as academic performance, understanding, and skill development.
3. To examine the link between student engagement and learning outcomes when AI is used in education.
4. To analyze students' dependence on AI tools and its impact on their independent thinking and creativity.

5. Hypotheses

H1: Engagement

- H0₁: AI tools have no significant impact on students' engagement in learning.
- H1₁: AI tools have a significant positive impact on students' engagement in learning.

H2: Learning Outcomes

- H0₂: AI usage has no significant effect on students' learning outcomes.
- H1₂: AI usage significantly improves students' learning outcomes.

H3: Dependency & Thinking Ability

- H0₃: AI usage does not lead to dependency or affect independent thinking and creativity.
- H1₃: AI usage leads to dependency and negatively affects independent thinking and creativity.

6. Research Methodology

The study employs a quantitative, descriptive, and analytical design to examine the impact of artificial intelligence (AI) on students' engagement, learning outcomes, and dependency.



Primary data was collected from 110 college students in Bengaluru using a structured questionnaire based on a five-point Likert scale.

Convenience sampling was used to gather responses across various academic streams.

Data analysis included descriptive statistics such as percentages and frequency distribution, along with inferential techniques like Chi-square tests and correlation analysis to test hypotheses and relationships.

The study maintains validity through alignment with objectives but is limited by a small sample size and reliance on self-reported data.

Demographic Summary Table

Variab le	Category	Frequen cy	Percenta ge
Gender	Male	36	34.30%
	Female	69	65.70%
Age Group	Below 18	10	9.50%
	18–19	68	64.70%
	20–21	22	21.00%
	Above 21	5	4.80%
Course / Stream	Commerce / /	75	71.40%

	Manageme nt		
	Science	20	19.00%
	Others (Arts, Nursing, etc.)	10	9.60%
Total Sample Size		105	100%

8. Analysis & Statistical Findings

The data collected from 110 respondents was analyzed using both descriptive and inferential statistical techniques to understand the impact of artificial intelligence (AI) on students' engagement, learning outcomes, and dependency.

8.1 Descriptive Analysis

The demographic profile indicates that the majority of respondents belong to the age group of 18–19 years, representing typical college-level students. Female respondents constituted a higher proportion compared to male respondents.



Most participants were from commerce and management backgrounds, particularly BCom and BBA, indicating relevance to academic usage of AI tools.

The analysis of Likert-scale responses reveals that a significant majority of students agree or strongly agree that AI tools enhance engagement, improve understanding of academic concepts, and positively influence academic performance.

Students also reported that AI supports skill development and helps in better retention of information.

However, responses regarding the reliability of AI tools showed moderate variation, indicating partial trust among users.

At the same time, a notable proportion of respondents agreed that AI usage leads to dependency and reduced independent thinking, highlighting a potential negative impact on creativity.

8.2 Likert-Scale Analysis

Statement	Mean	Std. Deviation	Interpretation
AI tools make my learning	3.85	0.78	Agree

more engaging and interactive			
I participate more actively in studies when using AI tools	3.8	0.81	Agree
AI helps me understand academic concepts more effectively	3.92	0.75	Agree
AI improves my academic performance and assignment quality	3.88	0.77	Agree
AI enhances my problem-solving and skill development abilities	3.86	0.79	Agree
When I am more engaged using AI, my learning outcomes improve	3.95	0.73	Agree



AI helps me retain information for a longer period of time	3.78	0.82	Agree
AI tools are reliable and provide accurate academic information	3.83	0.8	Agree
Use of AI has reduced my creative thinking ability	3.7	0.85	Neutral/Agree
I sometimes rely on AI instead of thinking independently	3.93	0.76	Agree

8.3 Inferential Analysis

The Chi-square test was used to examine the relationship between gender and students' perception of AI engagement. The results indicated no significant association, suggesting that perceptions of AI are consistent across genders.

Further, correlation analysis revealed a positive relationship between student engagement and learning outcomes. Students who reported higher engagement with AI tools also indicated better understanding, performance, and retention, leading to the acceptance of the alternative hypothesis.

Additionally, the analysis of responses related to dependency and creativity showed a consistent trend where increased reliance on AI corresponded with reduced independent thinking. This supports the hypothesis that AI usage contributes to dependency among students.

8.4 Correlation Analysis

Variables Compared	Correlation (r)	Strength
AI Usage ↔ Engagement	0.56	Moderate
AI Usage ↔ Academic Performance	0.48	Moderate
AI Usage ↔ Skill Development	0.36	Low



Engagement ↔ Learning Outcomes	0.52	Moderate
AI Usage ↔ Dependency	0.60	Strong
AI Usage ↔ Creativity Reduction	0.42	Moderate

Results & Discussion

The data collected from respondents was analyzed to examine the influence of Artificial Intelligence (AI) on students' engagement, academic performance, learning outcomes, and dependency on AI-based educational tools.

The findings indicate that AI plays a significant role in improving students' learning experiences by facilitating interactive learning, enhancing understanding of academic concepts, and supporting academic performance.

The analysis further reveals that students generally perceive AI tools as beneficial for improving their engagement and participation in educational activities. AI-

based applications provide personalized learning support, instant feedback, and assistance in understanding complex concepts, thereby contributing to improved learning efficiency.

The study also identifies a positive relationship between student engagement and learning outcomes. Students who reported higher engagement with AI tools also indicated better understanding, academic performance, and retention of information.

However, the findings highlight certain concerns associated with excessive dependence on AI tools. A notable proportion of respondents indicated that frequent AI usage may reduce their independent thinking and creative abilities. This suggests that while AI enhances academic support, overdependence may affect students' ability to develop original ideas and solve problems independently.

The analysis of gender-based perceptions indicates no significant association between gender and students' perceptions of AI-enabled learning engagement, suggesting that AI adoption is perceived relatively consistently across gender groups.



Overall, the findings demonstrate that AI positively contributes to student engagement and learning outcomes while simultaneously presenting challenges related to dependency and independent thinking. Therefore, the integration of AI into educational practices should emphasize balanced usage, critical thinking, and independent learning.

9. Conclusion

The study concludes that Artificial Intelligence has a significant influence on students' engagement and learning outcomes in educational institutions in Bengaluru.

The findings reveal that AI-based educational tools enhance student participation, improve understanding of academic concepts, support academic performance, and facilitate better retention of information.

The study also establishes a positive relationship between student engagement and learning outcomes, indicating that students who actively engage with AI-supported learning tools tend to report better academic experiences.

However, the study identifies increasing dependency on AI tools as a major concern. Excessive reliance on AI may negatively

influence students' independent thinking, creativity, and problem-solving abilities.

The findings emphasize that although AI provides substantial educational benefits, it should not replace independent intellectual effort or traditional learning processes.

Therefore, educational institutions should encourage the responsible and balanced integration of AI into teaching and learning practices. Students should be motivated to use AI as a supplementary learning resource while maintaining their ability to think critically, develop original ideas, and solve academic problems independently.

In conclusion, AI has considerable potential to transform modern education by improving engagement and learning effectiveness. Its successful integration, however, depends on maintaining an appropriate balance between technological assistance and independent student learning.

1, FUNDING STATEMENTS:

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors

2. Conflict of Interest Statement:



The authors declare no conflict of interest with respect to the research, authorship, or publication of this article.

3. Ethics Approval:

Ethics approval was not required for this study as it did not involve human/animal subjects.

4. Informed Consent:

Yes, informed written consent was obtained from all participants.

5. Data Availability Statement:

The data that support the findings of this study are available upon reasonable request from the corresponding author.

6. Author Contributions (Credit Statement):

☐ Author 1: Conceptualisation, Methodology, Writing – Original Draft

☐ Author 2: Data Curation, Formal Analysis, Writing – Review; Editing

BIBLIOGRAPHY

References

1. World Economic Forum. (2022). *The future of jobs report 2022*. World Economic Forum.

2. Manyika, J., et al. (2023). *Defining the skills citizens will need in the future world of work*. McKinsey Global Institute.

3. Singh, A., & Dey, S. (2022). *Skill development and employability in India: A study under NEP 2020*. *Journal of Education and Practice*, 13(4), 45–52.

4. Gratton, L. (2023). *The shift towards continuous learning in organizations*. Learning Press.

5. Chakraborty, R., & Misra, P. (2024). *Organizational learning culture and innovation performance*. *International Journal of Human Resource Studies*, 14(2), 88–102.

6. IBM. (2021). *Artificial intelligence in education: Enhancing learning experiences*. <https://www.ibm.com>

7. UNESCO. (2021). *AI and education: Guidance for policy-makers*. UNESCO Publishing.

8. Google. (2023). *AI tools in education and their impact on learning*. <https://www.google.com>

9. Statista. (2023). *Usage of AI tools among students worldwide*. <https://www.statista.com>

10. OpenAI. (2023). *ChatGPT and AI in education: Opportunities and challenges*. <https://www.openai.com>